Time Series and ARIMA

DSC 340 Applied Machine Learning

EMC Ch. 8

The Plan

- Introduce time series
- ARIMA model
  - We are omitting many technical details
- Other methods

Time Series

- **Discrete time**
  - Measurements are "evenly spaced" (time $t = 0, 1, 2, \ldots$)
  - Common examples: daily, monthly, every second
  - **Usually y_t is NOT independent of y_{t-1}**
  - To think about: **missing observations** (e.g., markets closed)
- Objectives
  - **Understand** the pattern (**infer** and **model** the structure)
  - **Forecast** (predict future values; extrapolating)
- Applications
  - Retail sales, inventory management, stock trading



George Box

(Quote: "all models are wrong")

(1919-2013)

https://commons.wikimedia.org/wiki/File:GeorgeEPBox.jpg



Gwilym Jenkins

Welsh statistician and systems engineer

(1932-1982)

https://alchetron.com/Gwilym-Jenkins

Box-Jenkins Methodology

- **Trend**
  - Long-term movement up or down ("secular trend")
  - Example: real estate values gradually increase in some market
- **Seasonality**
  - Examples: swimsuits and ski jackets; vacation rentals
- **Cyclic**
  - Similar to seasonal, but not tied to specific dates
  - Oil industry: "boom and bust"
- **Random**
  - "White noise"; often assume normally distributed

Fitting with Box-Jenkins

1. Condition the data and select a model
   - Identify and account for trends or seasonality
   - Examine the adjusted time series and determine suitable model
2. Estimate model parameters (fit model)
3. Assess the model (go to step 1 if necessary)
   - Note that since a sequence, we can't use *randomized* train-test split
   - Common to partition into two *sequences* to test how well it will generalize



https://otexts.com/fpp2/accuracy.html

ARIMA

- `AR` **Auto**regressive (auto = self)
- `I` Integrated (seasonal model)
- `MA` Moving Average

- Really a **hybrid** model (AR + MA = ARMA, also ARIMA)
- Often incorporates several different models (not just these), e.g.,
  - Long-term trend
  - Seasonal patterns
  - Vectorized (multi features)

Assumptions

- Stationary
  - Consistent in the time under observation (e.g., probabilities don't change)
  - Often becomes stationary after correcting for long-term and seasonal trends
- Current events only depend on a fixed number of past events
  - Markov assumption
  - "Of order p" (or q, etc.)
- Error term ε_t
  - Usually assume $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ for all t
  - White noise process
  - Random, independent fluctuations

Autoregressive Model

- AR(p): Read as "autoregressive model of order p"

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

- y_t is the value at time t
- ϕ_k is a constant for $k = 1, 2, \ldots, p$
- δ is some constant (informally, baseline or average)
- $\phi_p \neq 0$
- $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ for all t

Moving Average Model

- MA(q): Read as "moving average model of order q"

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

- y_t is the value at time t
- θ_k is a constant for $k = 1, 2, \ldots, q$
- $\theta_q \neq 0$
- $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ for all t

ARMA Model

- Combined model
- ARMA = AR(p) + MA(q)

$$y_t = \delta + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Trends in MA models

- Stationary assumption assumes no trend
- Often there is a trend, so we have to "correct" for this
- One approach: Fit linear (or higher order model) then subtract off
  - You can imagine this as "tilting" the model CW or CCW to make it flat
- More common approach: Differencing

$$d_t = y_t - y_{t-1} \text{ for } t = 2, 3, \dots, n$$

- Higher-order differencing ("twice differenced" or of order d)
  - But be careful: similar to fitting higher-order polynomials!
  - Can unnecessarily increase the variance

ARIMA Model

- Autoregressive **integrated** moving average model

$$\text{ARIMA}(p, d, q) \times (P, D, Q)_S$$

- p, d, q same as before AR(p), MA(q), d is # of times differenced
- s denotes seasonal period (e.g., 365 if annual, 7 if weekly, etc.)
- P # of terms in AR model across the s periods
- D # of differences applied across the s periods
- Q # of terms in the MA model across the s periods

Evaluating model

- Metrics such as AIC, AICc, BIC
- Normality and constant variance
  - Is the error actually normally distributed?
- How well does it extrapolate?
  - Example: train model for first 10 months
  - Evaluate model on the next 2 months of historical data
  - **Again: CAUTION! Extrapolation assumes stationarity!**
  - Long-term projections are particularly risky

Other considerations

- Not a causal model
  - Doesn't tells us how to improve our sales!
- Not resilient to severe shocks
- Most useful for short-term forecasting

Parameters and Hyperparameters

- What are the **parameters** and **hyperparameters**?
- Hyperparameters include: p, d, q
  - You also need to decide on timestep and seasonal period
  - But you probably make that decision even earlier, when preparing the data
- Parameters may include:
  - Coefficients and intercept for AR: $\phi_1, \dots, \phi_p$ and δ
  - Coefficients for MA: $\theta_1, \dots, \theta_q$
  - Differencing coefficients: $d_2, \dots, d_n$
  - There may be others: Remember, these are often complex, integrated models

Implementation

- Supported by R (see book)
- Also Jupyter/sklearn (see tutorial linked from Moodle)
- Many custom toolkits for stock trading, etc.

Other Methods

- Multivariate time series analysis (VARIMA = "vector ARIMA")
- Markov chains, hidden Markov models (HMMs), Kalman filters (state)
- Recurrent neural networks (RNNs): multi-layer perceptron with cycles
- Several other methods (see book, p. 253)